

Day 4: JavaScript Scraping & Other Roadblocks

Survey

- Apparently, I talked well-paced
- But some of you mentioned that the script was a bit messy — variables all over the place, and hard to navigate
 - I cannot fix that for today and tomorrow due to time constraints
 - But I will let Carl know & adjust this for next year!

Project / Office Hours

- Original plan: Have office hours on Friday
- But: That's a holiday
- So I'll be available Monday (May 4th), 9–11am
- Sign-up sheet on Lisam

Project

- Your first project is about scraping
- You have ~2wk
- Process:
 1. Find a website with interesting data
 2. Reflect on ethical & legal issues
 3. Use the tools from this week to download multiple pages and extract data from it
 4. Write it up
- I'll talk more about the dos and don'ts tomorrow, where you can also ask questions
- I want you to find (a) website(s) that look(s) promising
- If you want an estimate (is the project too large/too small), please come to the office hours
- Or with any other questions

Today's Agenda

- We're almost done!
- Now we focus on two final “end bosses”: Scraping JavaScript and Regular Expressions
 - Regular Expressions are on the menu for tomorrow.

Important

- What I am about to show you is **optional**.
- Dealing with JavaScript-based websites is hard, and getting increasingly harder
- Especially if you're new to coding, this can prove to be unobtainable
- That's why it's optional.



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#

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Aggy Wonx @aggrohwonx.bsky.social · 7m

the sociology major in me sees very terrifying patterns emerging, advantageous to fascist capitalists.

no one would believe me, however.



Aggy Wonx @aggrohwonx.bsky.social · 14m

Im glad my mom taught me basic critical literacy thinking cause fuck man i cannot IMAGINE raw dogging social media, there are some fucking broken people cause of these algorithms man



1



RNS @rns.org · 11m

OPINION: (The Conversation) — Long-standing Christian narratives about Islam continue to inform political language, argues a sociologist of Islam.



The enduring legacy of medieval Christian depictions of Islam in today's political discourse

(The Conversation) — Long-standing Christian narratives about Islam continue to inform political language, argues a sociologist of Islam.

religionnews.com



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Aggy Wonx @aggrohwonx.bsky.social · 8m

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- Back
- Forward
- Reload
- Bookmark Page...
- Save Page As...
- Select All
- Take Screenshot
- View Page Source**
- Inspect Accessibility Properties
- Inspect
- Send to Device
- Flagfox
- Block element...
- Zotero Connector

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hwonx.bsky.social · 15m

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```

Note also how they forgot to close this tag... 🤔

The Basic Idea Behind SPAs

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Single-Page-Application</title>
  <script src="/app.js"></script>
</head>
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The Basic Idea Behind SPAs

How can you scrape a website using R that requires JavaScript?

The Basic Idea Behind SPAs

How can you scrape a website using R that requires JavaScript?

Tell R to use an actual browser.

Scraping JavaScript

Puppeteering your way through CAPTCHAs

Scrapping JavaScript

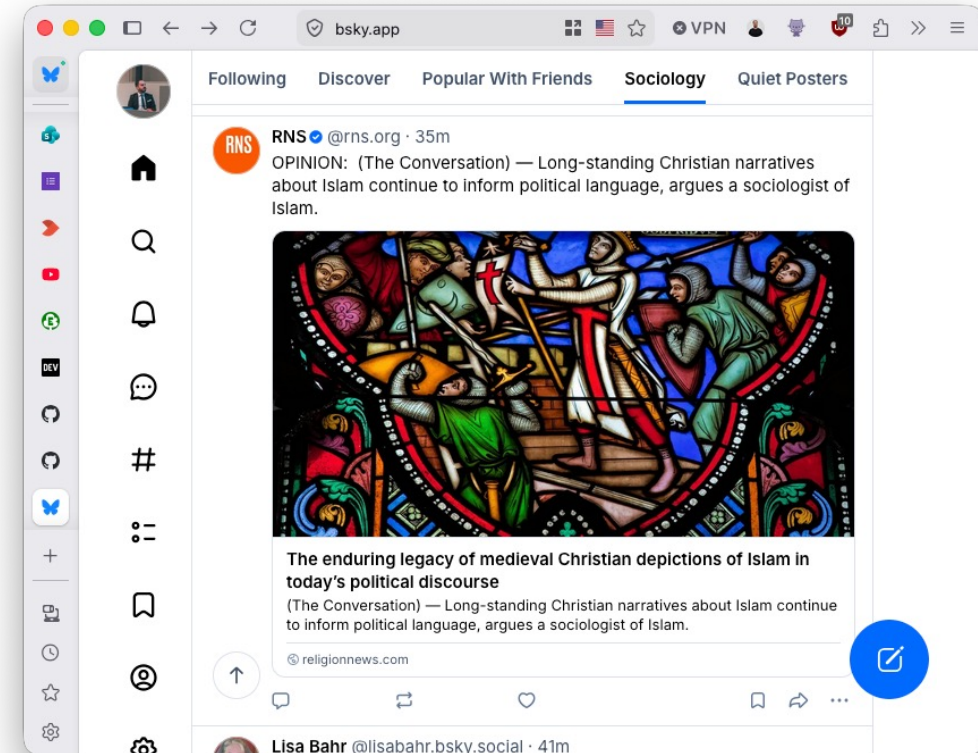
- Many websites are implemented as so-called “SPAs” (Single-Page Applications)
- The actual HTML that the web server gives you is a small skeleton
- It contains a `<script src="">` that links to a JavaScript file
- Your browser executes this and that's what actually loads all the data

Scraping JavaScript

- To scrape SPAs, we need to be able to execute JS
- Normal scraping will not do this.
 - → Solution: Selenium

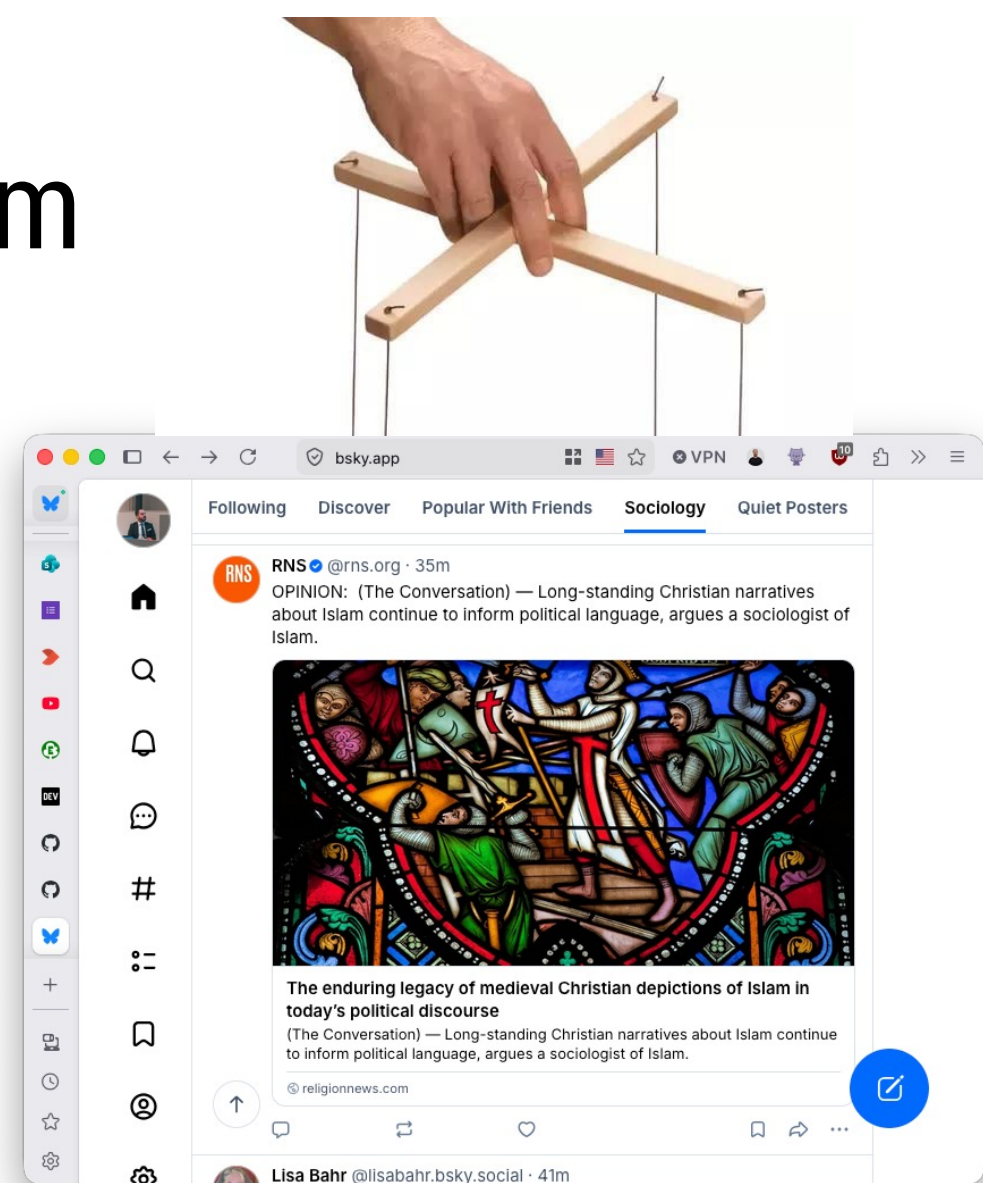
Introduction to Selenium

- Goal: Let R use an actual, real browser that executes JavaScript and thus lets JavaScript-based websites render properly



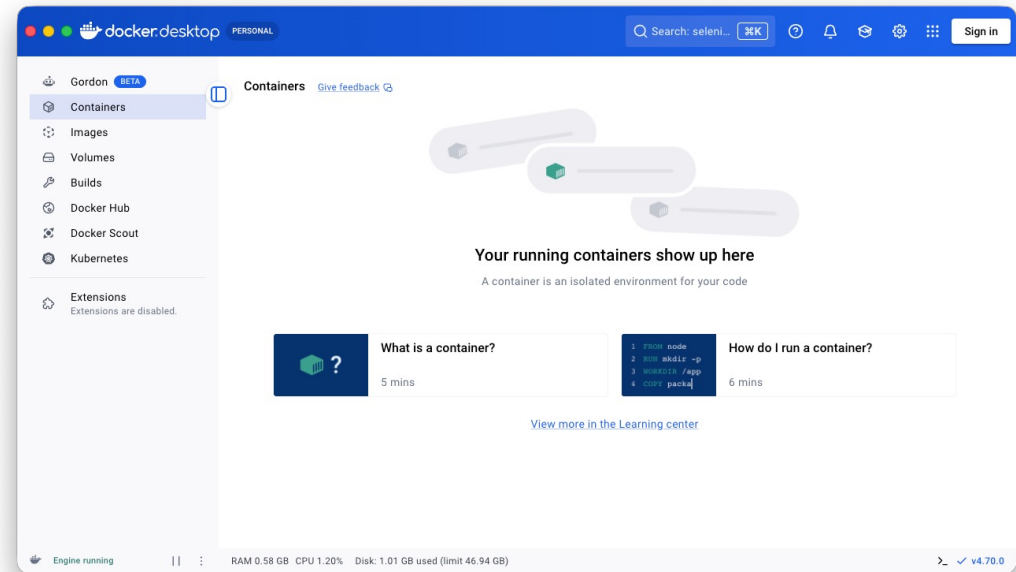
Introduction to Selenium

- Goal: Let R use an actual, real browser that executes JavaScript and thus lets JavaScript-based websites render properly
- For this to work, we need *another* piece of software that controls the browser → called a “driver”
- Selenium is such a driver



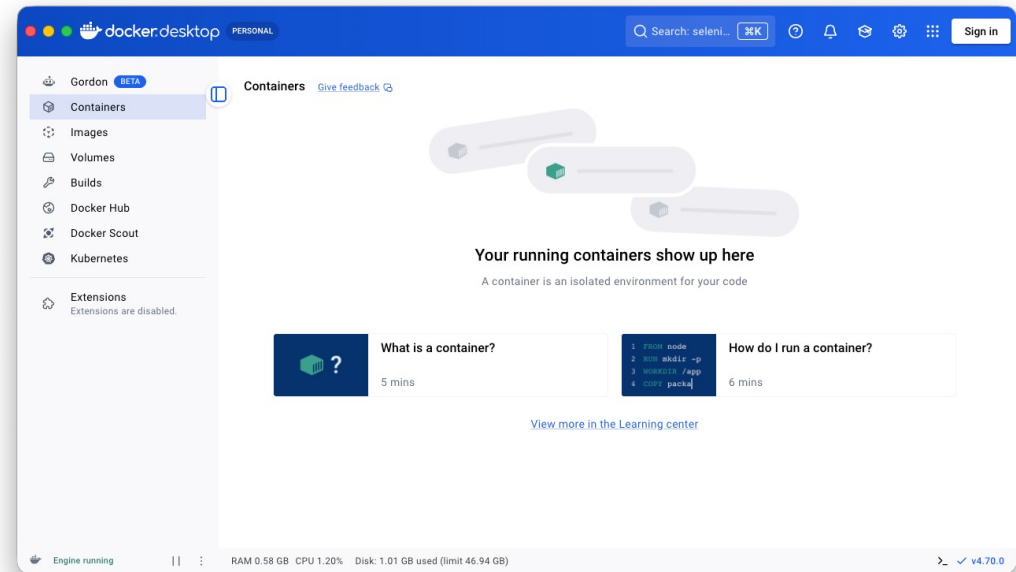
Introduction to Selenium

- Because Selenium is built with Java, installing it can be a bit difficult
- I personally recommend going the “Docker” route

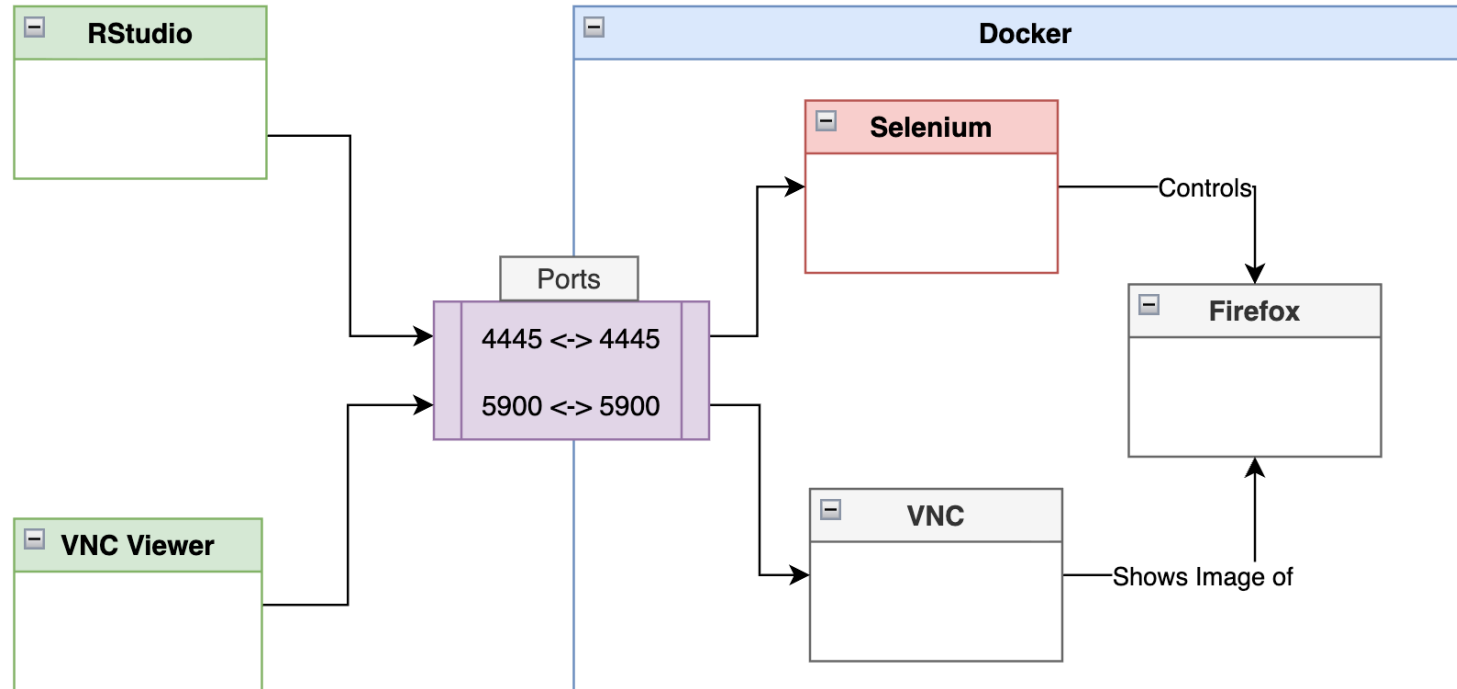


Introduction to Selenium

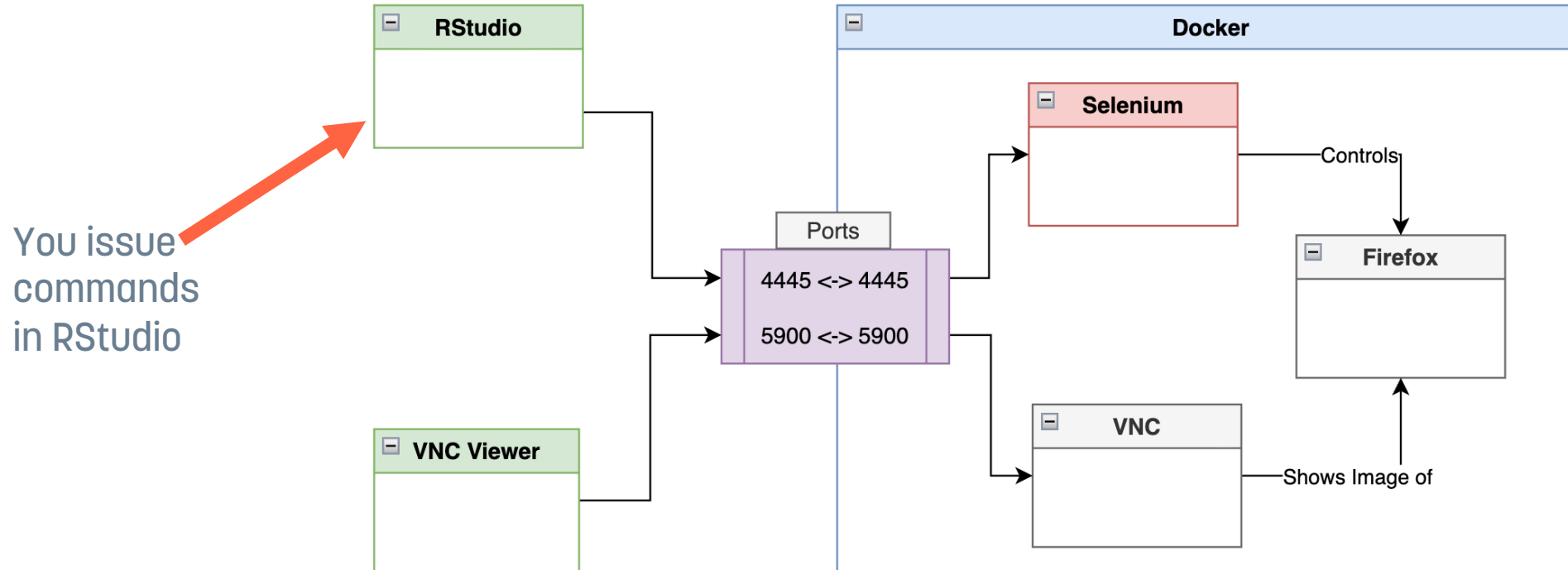
- Docker is a piece of software that allows you to run virtual machines on your computer
- How it works:
 1. Install Docker on your computer
 2. Then, download the Selenium package
 3. Run it and control it with R



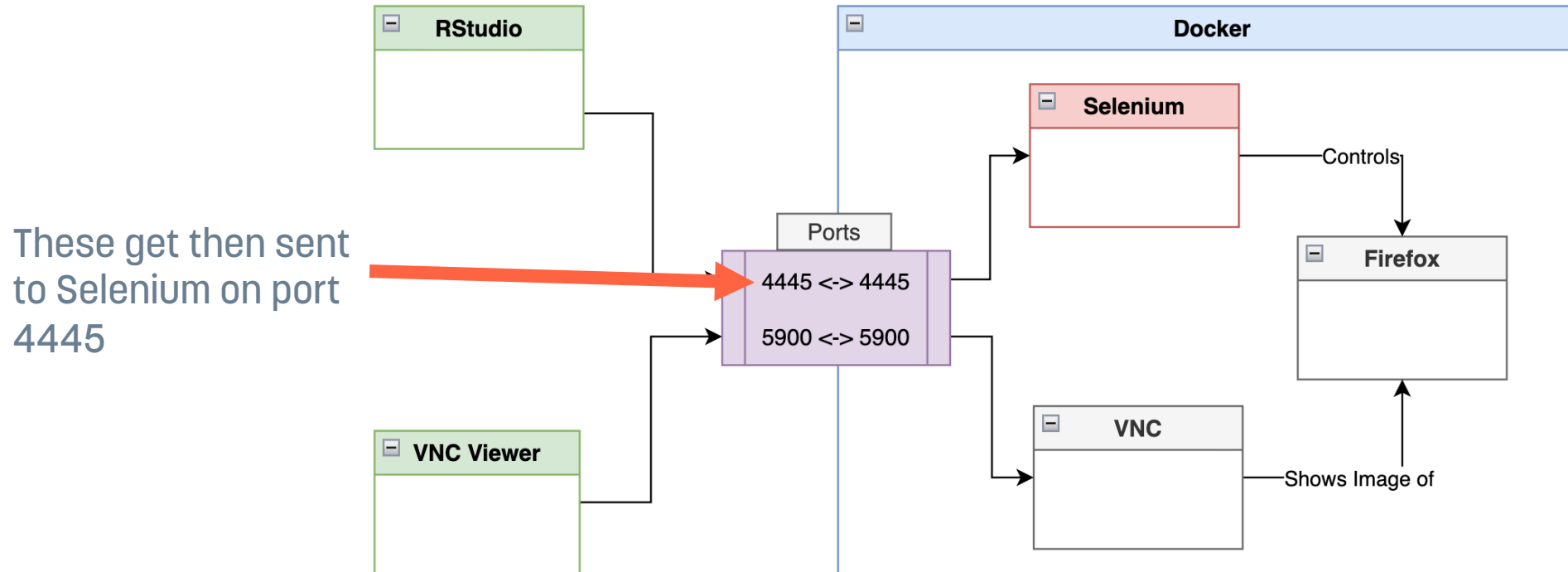
Introduction to Selenium



Introduction to Selenium

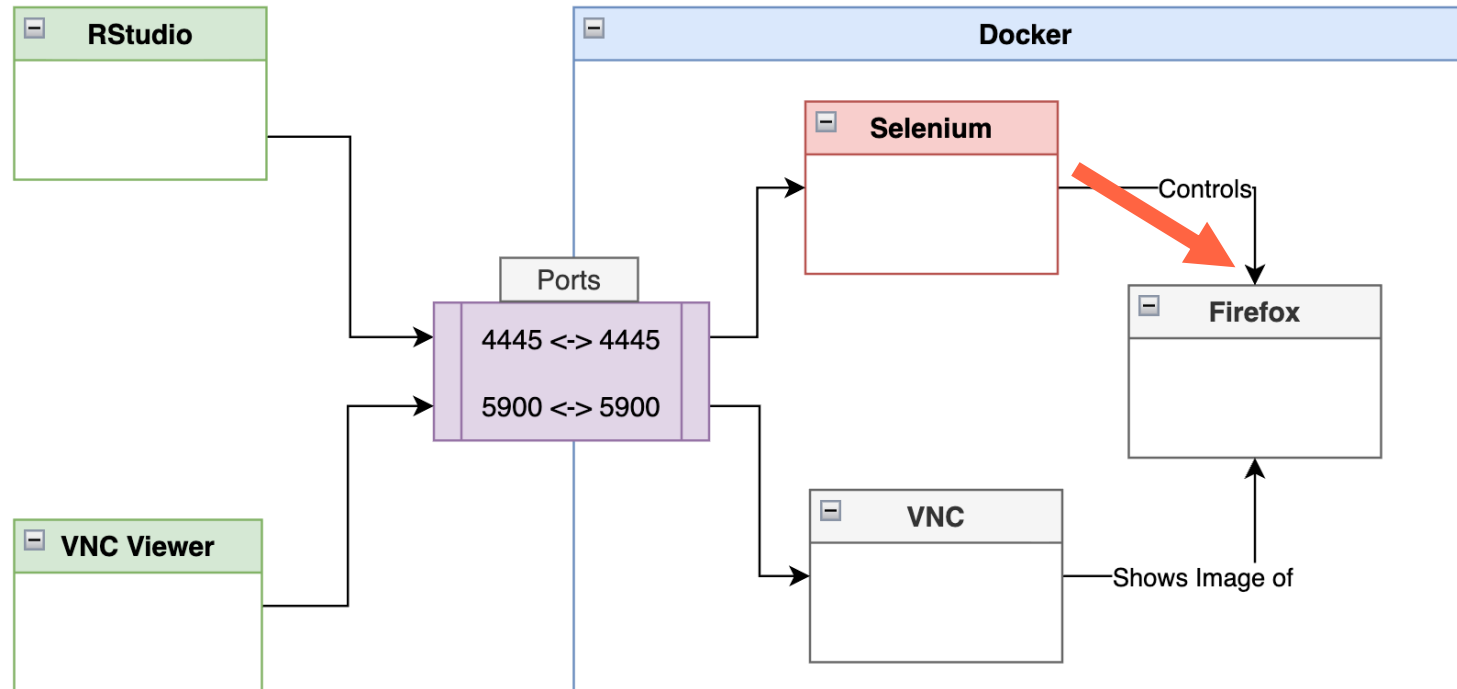


Introduction to Selenium



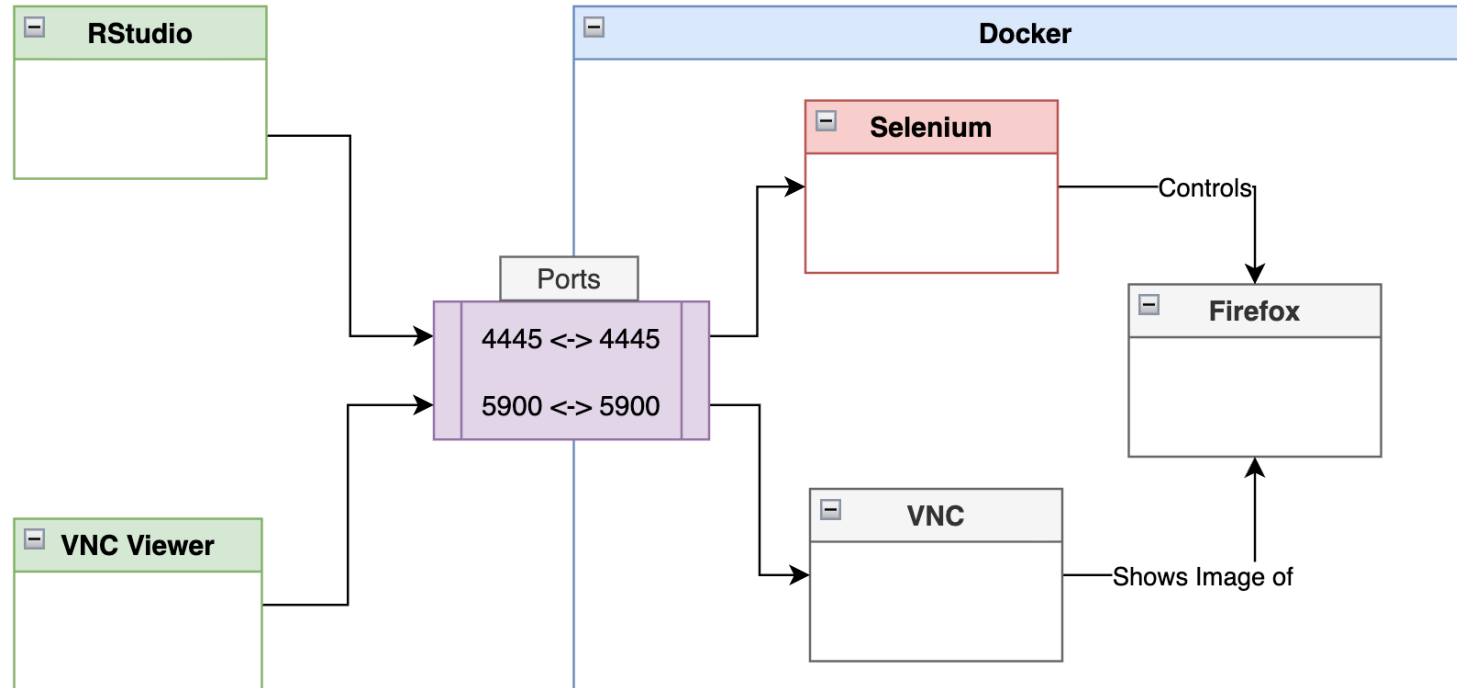
Introduction to Selenium

Selenium does
what you request in
its own Firefox



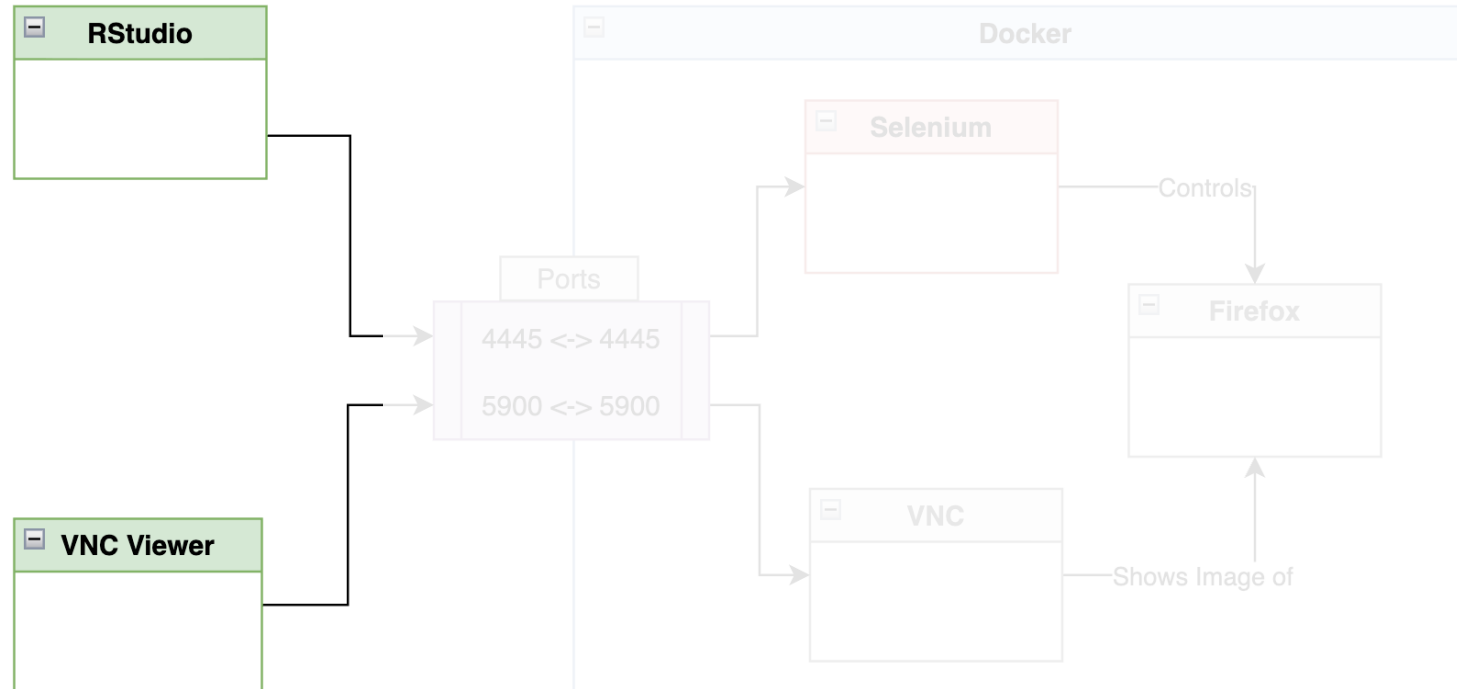
Introduction to Selenium

Using a “VNC” viewer you can also observe what is happening, and click on elements yourself.



Introduction to Selenium

Once installed, try
to forget what is
going on under the
hood



Introduction to Selenium

```
session <- SeleniumSession$new(browser = "firefox")

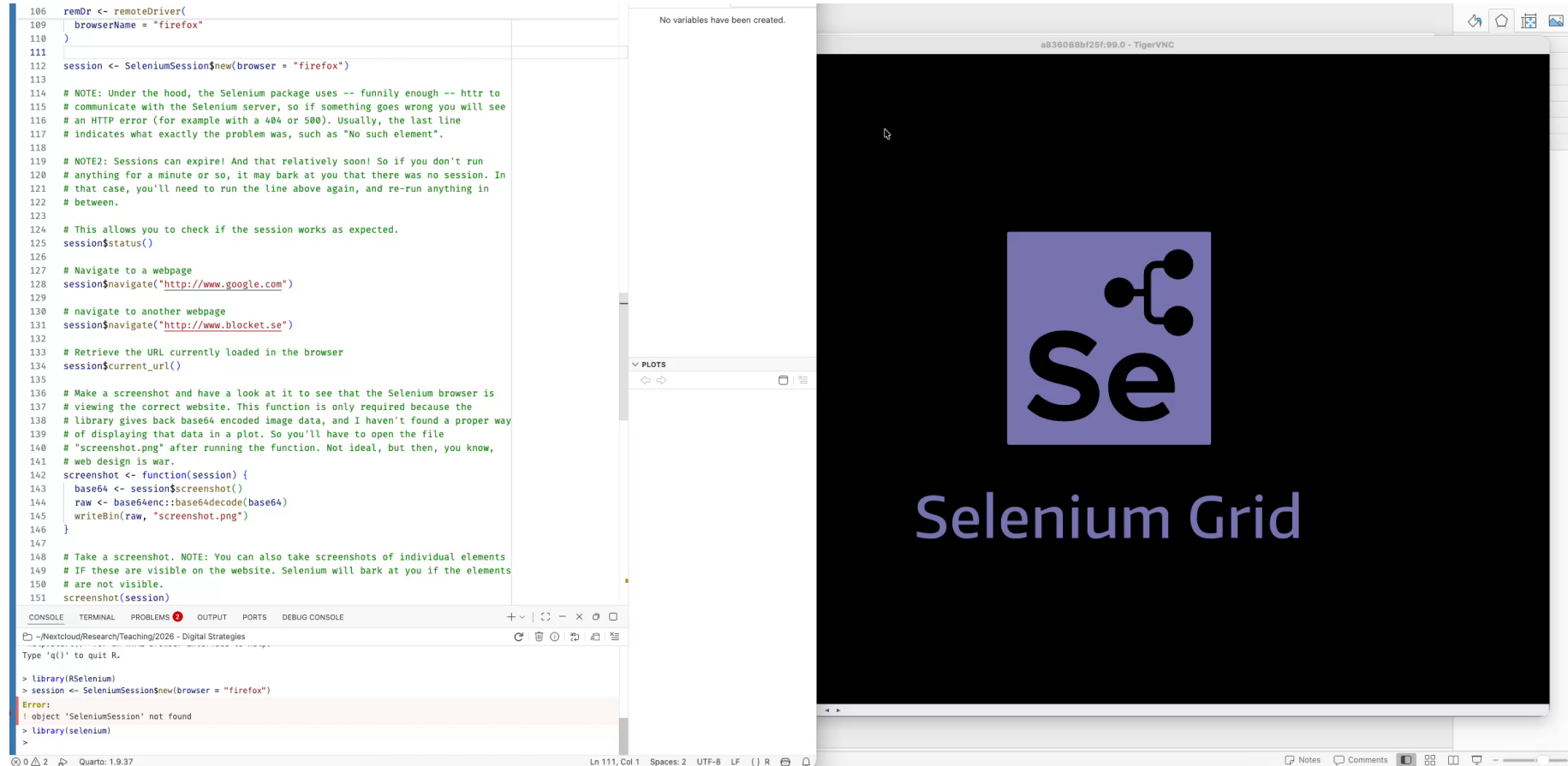
# First, navigate to the website:
session$navigate("https://soziologieblog.hypotheses.org/")

# Observe what happens: Anubis detects that the client accessing the website is possibly a bot and
# provides a much harder challenge than if you visited using your regular browser. It will take some
# time to do the challenge, but it will eventually work, and you can see the website.

# Next, there is a Cookie consent banner. Let's find and close it.
# By inspecting the "Accept all" (German: "Alle akzeptieren") button in the DevTools, you can
# see what you need to find the button.
button <- session$find_element(using="xpath", value="//a[contains(text(), 'Alle akzeptieren')]")

# Now we can programmatically "click" it to close the banner:
button$click()
```

Introduction to Selenium



The image shows a screenshot of an RStudio/Quarto environment. The main window displays R code for setting up a Selenium session. The code includes comments about the Selenium package's use of HTTP, session expiration, and a function to take screenshots. The console shows an error message: "Error: object 'SeleniumSession' not found". A small plot window titled "PLOTS" is visible, showing the Selenium logo and the text "Selenium Grid". The status bar at the bottom indicates "Ln 111, Col 1 Spaces: 2 UTF-8 LF () R".

```
106 remDr <- remoteDriver(  
107   browserName = "firefox"  
108 )  
109  
110 session <- SeleniumSession$new(browser = "firefox")  
111  
112 # NOTE: Under the hood, the Selenium package uses -- funnily enough -- http to  
113 # communicate with the Selenium server, so if something goes wrong you will see  
114 # an HTTP error (for example with a 404 or 500). Usually, the last line  
115 # indicates what exactly the problem was, such as "No such element".  
116  
117 # NOTE2: Sessions can expire! And that relatively soon! So if you don't run  
118 # anything for a minute or so, it may bark at you that there was no session. In  
119 # that case, you'll need to run the line above again, and re-run anything in  
120 # between.  
121  
122 # This allows you to check if the session works as expected.  
123 session$status()  
124  
125 # Navigate to a webpage  
126 session$navigate("http://www.google.com")  
127  
128 # navigate to another webpage  
129 session$navigate("http://www.blocket.se")  
130  
131 # Retrieve the URL currently loaded in the browser  
132 session$current_url()  
133  
134 # Make a screenshot and have a look at it to see that the Selenium browser is  
135 # viewing the correct website. This function is only required because the  
136 # library gives back base64 encoded image data, and I haven't found a proper way  
137 # of displaying that data in a plot. So you'll have to open the file  
138 # "screenshot.png" after running the function. Not ideal, but then, you know,  
139 # web design is war.  
140 screenshot <- function(session) {  
141   base64 <- session$screenshot()  
142   raw <- base64enc::base64decode(base64)  
143   writeBin(raw, "screenshot.png")  
144 }  
145  
146 # Take a screenshot. NOTE: You can also take screenshots of individual elements  
147 # IF these are visible on the website. Selenium will bark at you if the elements  
148 # are not visible.  
149 screenshot(session)  
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CONSOLE

```
~/Nextcloud/Research/Teaching/2026 - Digital Strategies  
Type 'q()' to quit R.  
  
> library(RSelenium)  
> session <- SeleniumSession$new(browser = "firefox")  
Error:  
! object 'SeleniumSession' not found  
> library(selenium)  
>
```

Ln 111, Col 1 Spaces: 2 UTF-8 LF () R

Notes Comments

Some more Tricks

to make your web scraping endurable

Some more Tricks

- Étienne discussed on today (the fourth day) automation
- We already did that!
- So instead today is more of what you already did!
 - → Nothing new, just more exercises
- But this means I can also talk about a few other things

Handling Rate Limits

- Many websites do not check how often you visit them
- What do you do?

Handling Rate Limits

- Many websites do not check how often you visit them
- What do you do?
 - → use `sys.sleep(seconds)` to be nice to the website.

Handling Rate Limits

- Some enforce a “soft” limit on how often you can visit them
- That is called “rate limiting”
 - → it limits the rate at which you can request more data from the website

Handling Rate Limits

- Some enforce a “soft” limit on how often you can visit them
- That is called “rate limiting”
 - → it limits the rate at which you can request more data from the website
- If you visit such a website too fast, you will start getting 429 Too Many Requests errors

Handling Rate Limits

- You can detect rate limits by looking at the **response headers**

```
library(httr)
library(XML)

url <- "https://api.github.com/repos/Zettlr/Zettlr/releases"
response <- GET(url)
```

server	"github.com"
accept-ranges	"bytes"
x-ratelimit-limit	"60"
x-ratelimit-remaining	"58"
x-ratelimit-used	"2"
x-ratelimit-resource	"core"
x-ratelimit-reset	"1777210136"
x-github-request-id	"AE0C:83BEB:2FE154:42E780:69EE0F74"

Handling Rate Limits

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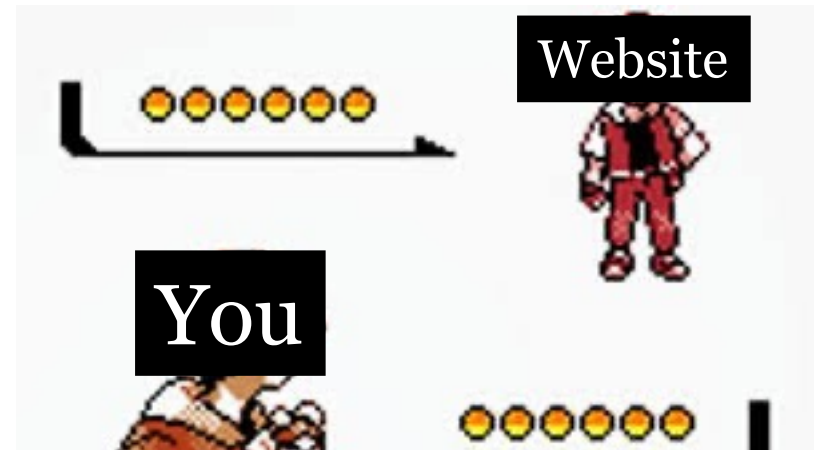
Important: These headers are not standardized, so different websites will tell you your remaining rate limit in different ways! → Always double-check!

Handling Blocks

- If you hit a rate limit, you will not be able to request any more data until the limit resets
- Some websites will be very harsh and completely lock you out

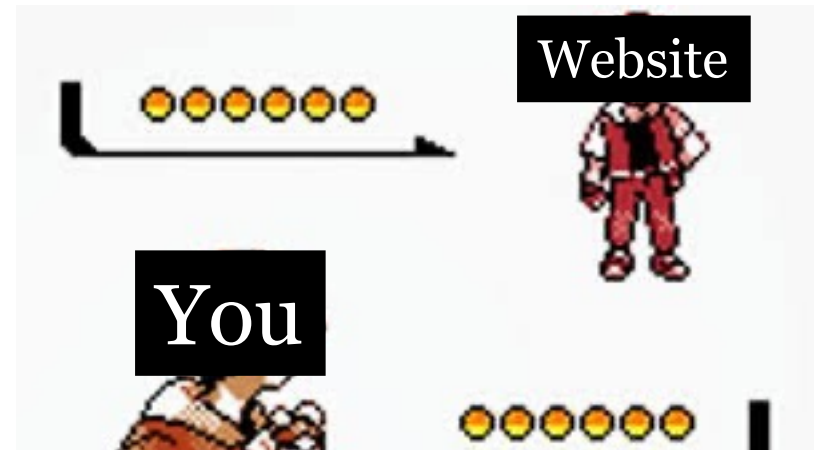
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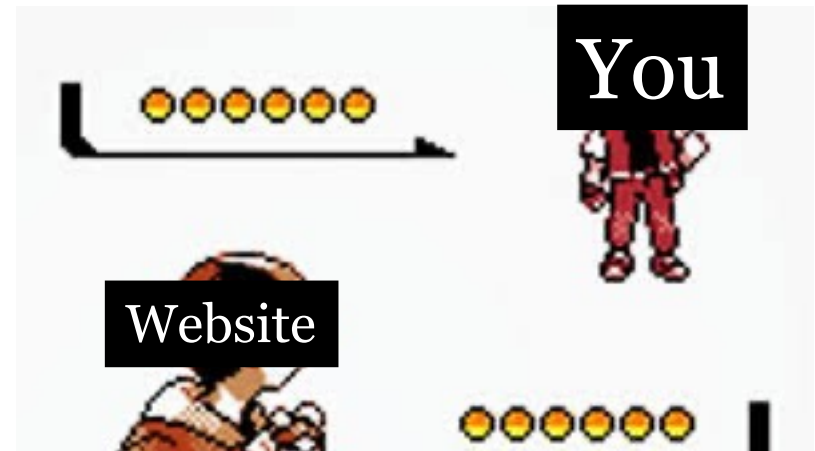
Handling Blocks

- If you hit a rate limit, you will not be able to request any more data until the limit resets
- Some websites will be very harsh and completely lock you out
- Let's switch perspectives for a moment



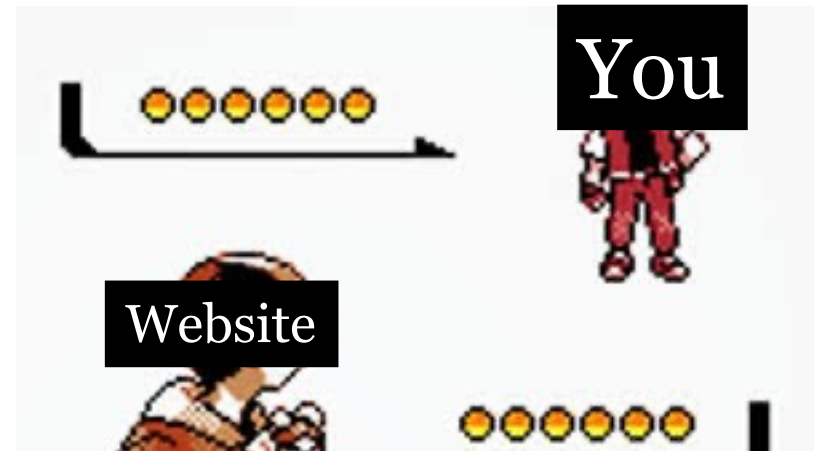
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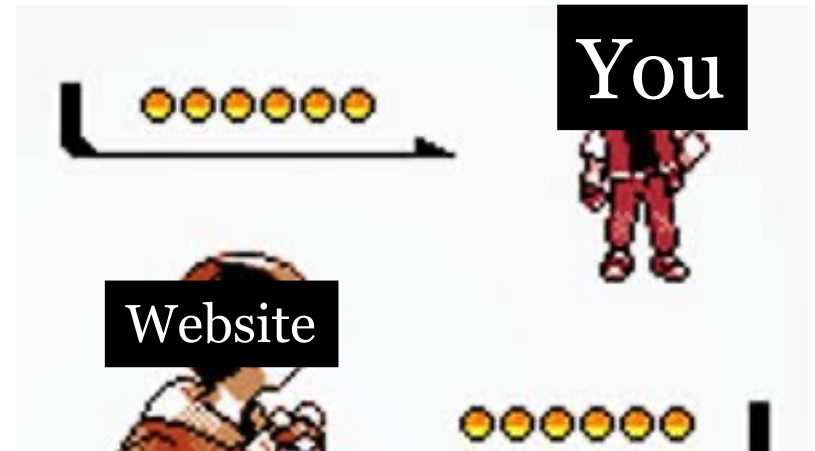
Handling Blocks

- We have talked about HTTP requests
- One thing I haven't talked about yet is that every request has an envelope
- Imagine mail: You send it somewhere (google.com), but you always have a sender address on it



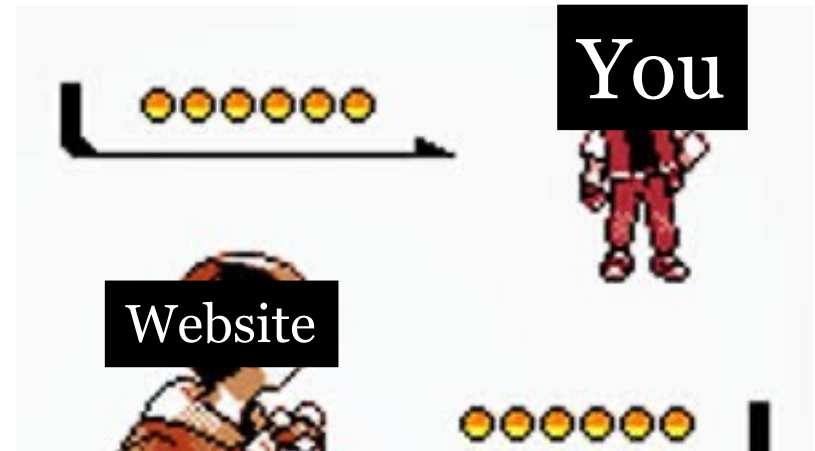
Handling Blocks

- Web servers will receive your IP address
- They may decide to place your IP address in a timeout



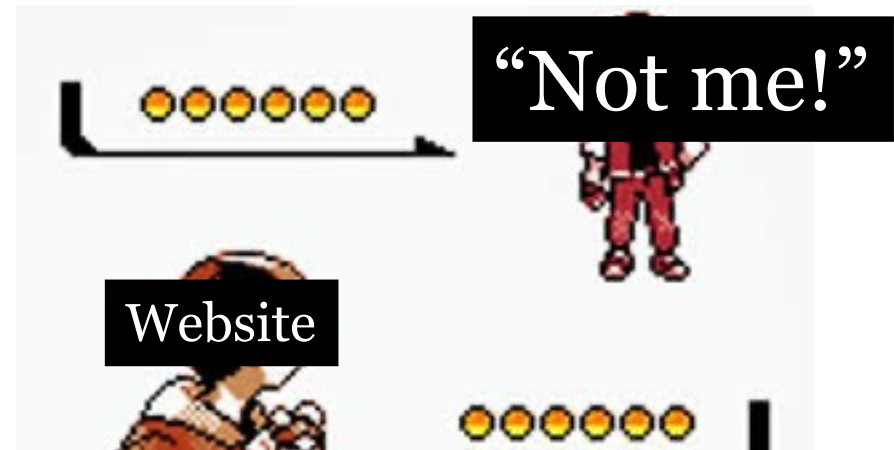
Handling Blocks

- Web servers will receive your IP address
- They may decide to place your IP address in a timeout
- What can you do?



Handling Blocks

- Web servers will receive your IP address
- They may decide to place your IP address in a timeout
- What can you do?
- Get a different IP, obviously!

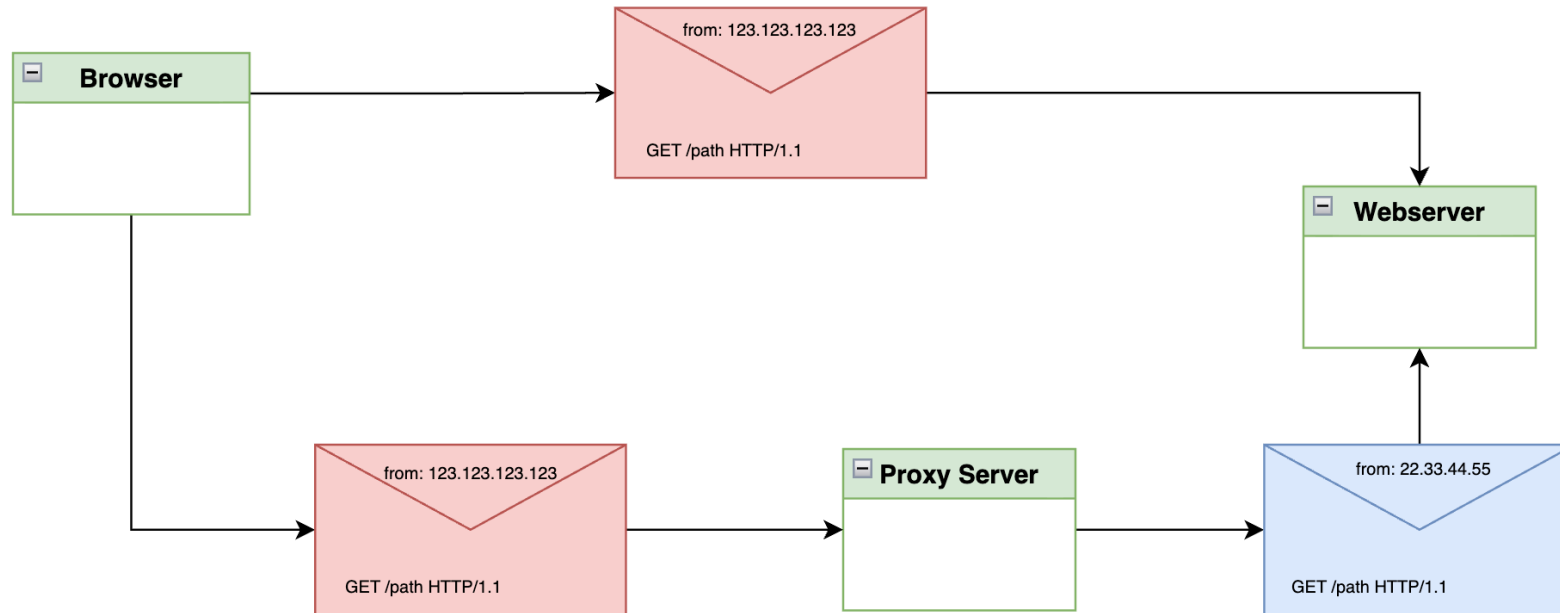


Proxy Servers

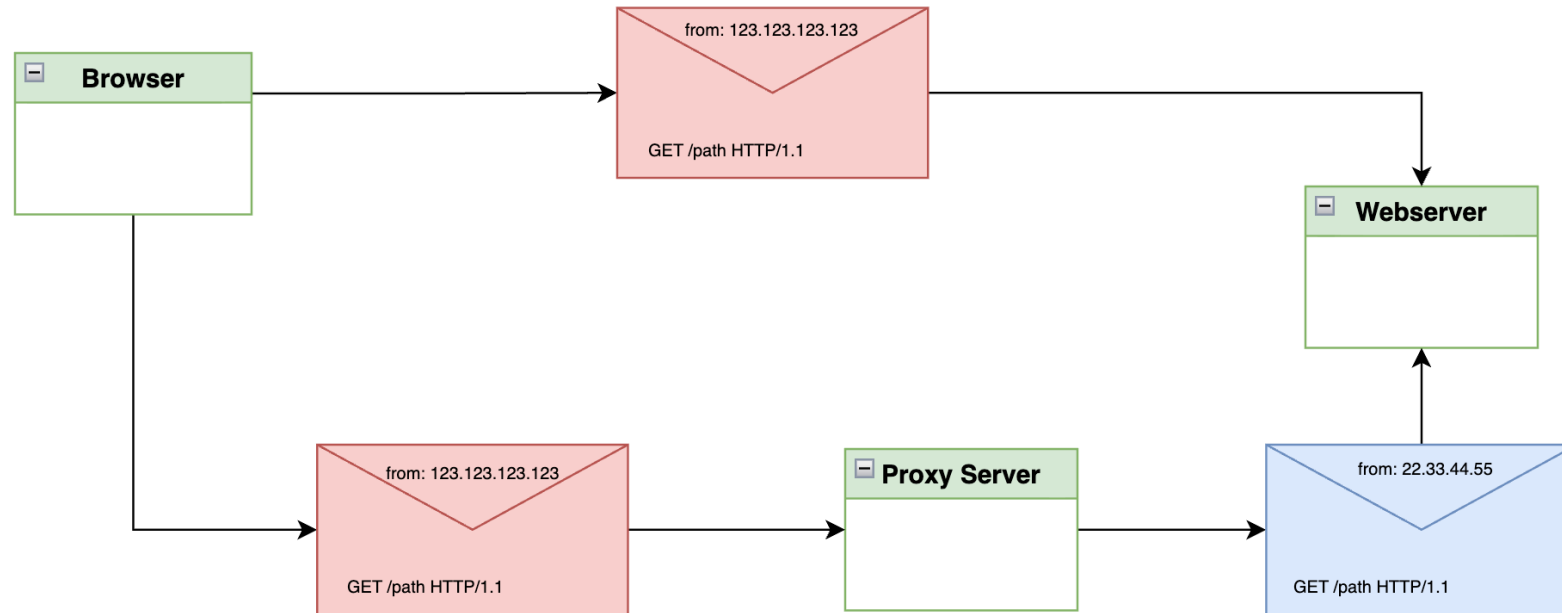
- A proxy server is yet another server
- Instead of handing out HTML, it allows you to hide your own IP address



Proxy Servers



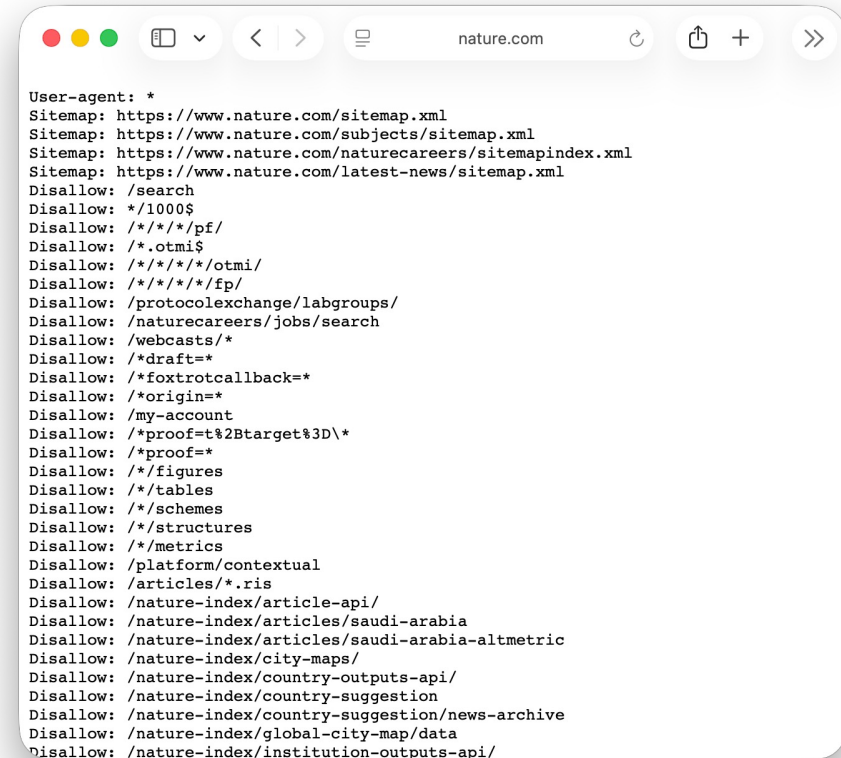
Proxy Servers



Very important: Double-check and reflect whether this is actually appropriate! Remember both the ethical and legal dimensions – ask yourself: Should I really do this?

robots.txt

- Many websites provide you with a file called robots.txt
- In this file, they explicitly tell you which parts you can scrape, and which one you should not
- It is not a legal requirement, it is more of a suggestion → but it can be *unethical* to scrape pages it tells you not to!

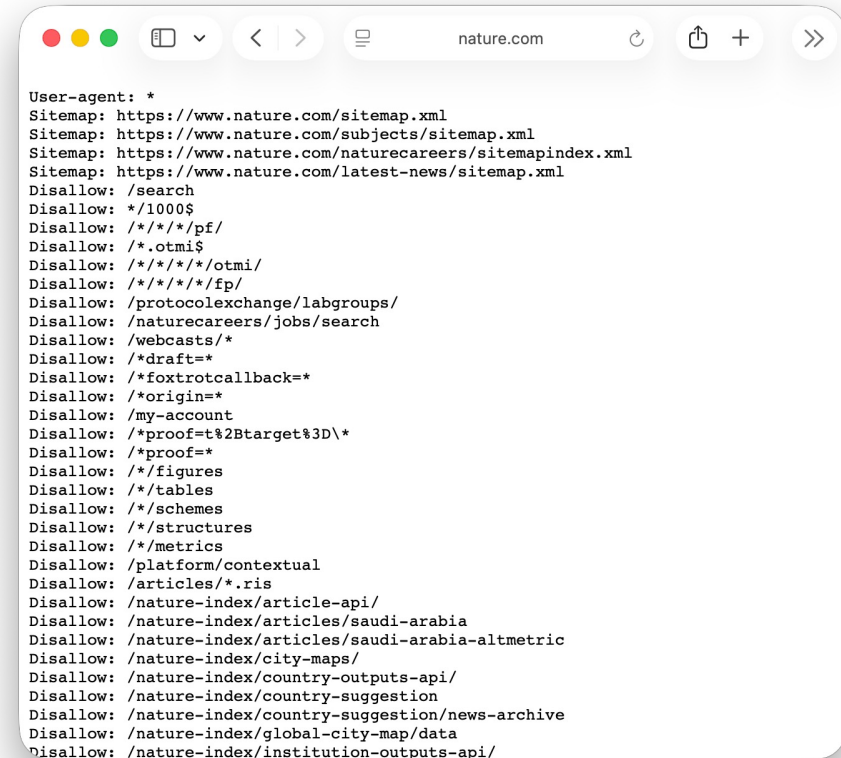


The image shows a web browser window with the address bar displaying "nature.com". The main content area shows the text of the robots.txt file for nature.com. The file contains several Sitemap and Disallow directives. The Sitemap directives point to various XML sitemaps on the nature.com domain. The Disallow directives list numerous paths and patterns that are disallowed for crawling, including search results, specific file types like .otmi, and various API endpoints and internal paths.

```
User-agent: *
Sitemap: https://www.nature.com/sitemap.xml
Sitemap: https://www.nature.com/subjects/sitemap.xml
Sitemap: https://www.nature.com/naturecareers/sitemapindex.xml
Sitemap: https://www.nature.com/latest-news/sitemap.xml
Disallow: /search
Disallow: */1000$
Disallow: /*/*/*pf/
Disallow: /*.otmi$
Disallow: /*/*/*/*otmi/
Disallow: /*/*/*/*fp/
Disallow: /protocolexchange/labgroups/
Disallow: /naturecareers/jobs/search
Disallow: /webcasts/*
Disallow: /*draft=*
Disallow: /*foxtrotcallback=*
Disallow: /*origin=*
Disallow: /my-account
Disallow: /*proof=t%2Btarget%3D\*
Disallow: /*proof=*
Disallow: /*/figures
Disallow: /*/tables
Disallow: /*/schemas
Disallow: /*/structures
Disallow: /*/metrics
Disallow: /platform/contextual
Disallow: /articles/*.ris
Disallow: /nature-index/article-api/
Disallow: /nature-index/articles/saudi-arabia
Disallow: /nature-index/articles/saudi-arabia-altmetric
Disallow: /nature-index/city-maps/
Disallow: /nature-index/country-outputs-api/
Disallow: /nature-index/country-suggestion
Disallow: /nature-index/country-suggestion/news-archive
Disallow: /nature-index/global-city-map/data
Disallow: /nature-index/institution-outputs-api/
```

robots.txt

- Will be located *always* at the root path `/robots.txt`
 - ex: `http://example.com/robots.txt`
- Structure:
 - User-agent: *
 - Sitemap: URL
 - Disallow: URL
 - Allow: URL

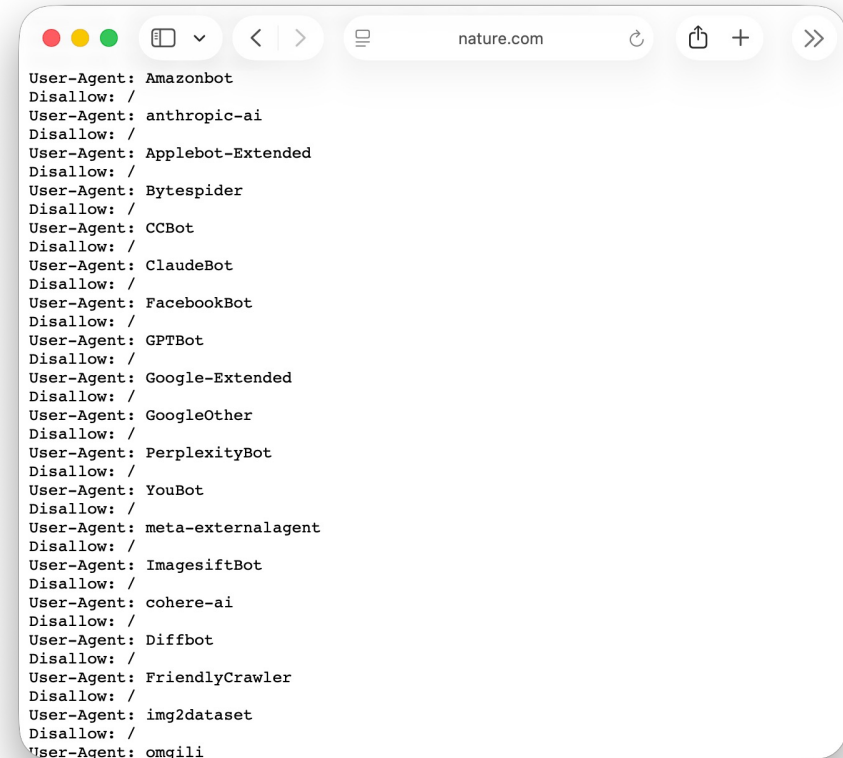


The screenshot shows a web browser window with the address bar displaying "nature.com". The main content area shows the text of the robots.txt file for nature.com. The file includes several Sitemap URLs and a list of disallowed paths.

```
User-agent: *
Sitemap: https://www.nature.com/sitemap.xml
Sitemap: https://www.nature.com/subjects/sitemap.xml
Sitemap: https://www.nature.com/naturecareers/sitemapindex.xml
Sitemap: https://www.nature.com/latest-news/sitemap.xml
Disallow: /search
Disallow: */1000$
Disallow: */**/pf/
Disallow: *.otmi$
Disallow: */**/*/*otmi/
Disallow: */**/*/*fp/
Disallow: /protocolexchange/labgroups/
Disallow: /naturecareers/jobs/search
Disallow: /webcasts/*
Disallow: /*draft=*
Disallow: /*foxtrotcallback=*
Disallow: /*origin=*
Disallow: /my-account
Disallow: /*proof=t%2Btarget%3D\*
Disallow: /*proof=*
Disallow: */figures
Disallow: */tables
Disallow: */schemas
Disallow: */structures
Disallow: */metrics
Disallow: /platform/contextual
Disallow: /articles/*.ris
Disallow: /nature-index/article-api/
Disallow: /nature-index/articles/saudi-arabia
Disallow: /nature-index/articles/saudi-arabia-altmetric
Disallow: /nature-index/city-maps/
Disallow: /nature-index/country-outputs-api/
Disallow: /nature-index/country-suggestion
Disallow: /nature-index/country-suggestion/news-archive
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robots.txt

- Will be located *always* at the root path `/robots.txt`
 - ex: `http://example.com/robots.txt`
- Structure:
 - `User-agent: *`
 - `Sitemap: URL`
 - `Disallow: URL`
 - `Allow: URL`



```
User-Agent: Amazonbot
Disallow: /
User-Agent: anthropic-ai
Disallow: /
User-Agent: Applebot-Extended
Disallow: /
User-Agent: Bytespider
Disallow: /
User-Agent: CCBot
Disallow: /
User-Agent: ClaudeBot
Disallow: /
User-Agent: FacebookBot
Disallow: /
User-Agent: GPTBot
Disallow: /
User-Agent: Google-Extended
Disallow: /
User-Agent: GoogleOther
Disallow: /
User-Agent: PerplexityBot
Disallow: /
User-Agent: YouBot
Disallow: /
User-Agent: meta-externalagent
Disallow: /
User-Agent: ImagesiftBot
Disallow: /
User-Agent: cohere-ai
Disallow: /
User-Agent: Diffbot
Disallow: /
User-Agent: FriendlyCrawler
Disallow: /
User-Agent: img2dataset
Disallow: /
User-Agent: omgili
```

Any Questions?

Thank you!